

CodeWeaver comparison with AlphaTrans

Four of AlphaTrans's ten exact Java projects were already measured in the published CodeWeaver matrix. All 12 translations compiled; none passed every available fixed runtime case. Across repetitions, 1130/3543 fixed cases passed. AlphaTrans's fragment-level TPR is preserved separately because it is not the same unit or denominator as CodeWeaver's project-test execution.

Scope

The exact common subjects are commons-cli, commons-csv, commons-fileupload, and commons-validator. Six paper subjects were not run, so this artifact makes no ten-project aggregate claim.

Unavailable surface

AlphaTrans's manual type-map completion, manual repair effort, GraalVM fragment validation, and human bug taxonomy cannot be recovered from CodeWeaver's end-to-end project outputs and are reported as unavailable rather than zero.

Exact-subject comparison

Project	AlphaTrans syntax	AlphaTrans TPR	CW builds	CW fixed tests
commons-cli	100.00%	10.08%	3/3	1103/1143 (96.50%)
commons-csv	98.72%	0.00%	3/3	0/894 (0.00%)
commons-fileupload	100.00%	63.44%	3/3	27/117 (23.08%)
commons-validator	99.23%	11.70%	3/3	0/1389 (0.00%)

CodeWeaver repetitions

Rep	Build	Pass all	Fixed tests	Test rate
1	4/4	0/4	379/1181	32.09%
2	4/4	0/4	373/1181	31.58%
3	4/4	0/4	378/1181	32.01%